Real-time movement correction in the gym

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Abstract

This paper investigates a deadlift correction system that gives visual feedback to the user aiming to reduce risk of injuries and improve the form while deadlifting. The system uses a side-view camera to capture the user’s form when performing a deadlift, which is then displayed on a screen with coloured lines indicating correct form and incorrect form. The screen is surrounded by LEDs which turn on when the user doesn't have the correct form when deadlifting. The system is a solution for males between the age of 18 - 25 that go to the gym, since they have the highest chance of injuring themselves in the lower back (Cuthbertson-Moon et al., 2024).

To understand the state of the art as well as to improve the acceptance of the design by the user group. A context analysis was conducted through a literature study and interviews. In which we found that users are interested in the product however the feedback style can influence the acceptance of the feedback. In addition, the group was also hesitant about the use of AI to correct their form when performing deadlifts. Based on these findings we designed a visual feedback system for the user's coloured lines. During the low fidelity testing we tested the intuitiveness of the design and found that the colours of the design had to be changed and that the visual feedback had too big of a delay. During the high fidelity test we found that the instruction video explaining the functionality of the system was needed for a quick understanding, that the LEDs needed to be less bright and the colours of the lines needed to be changed to make it more intuitive.

The system offers a unique, cost-effective solution to improve the form when performing a deadlift. Especially those who are training on their own. However, the system has a set of limitations such as system delay, camera sensitivity, and time constraints during development. These limitations could be an area for future improvement. Further testing is needed to assess the system’s effectiveness in reducing injuries and its acceptance in the market.

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Chapter 1: Introduction

Fundamentals for a healthy life are sports and movement. Being physically active can reduce the risk of heart attacks and dementia (Williamson, 2024). Sports can be done in teams or on an individual basis. Nowadays a lot of people are going to the gym, especially people in young adulthood (Upton-Clark, 2024). In the gym there is a big variety of exercise opportunities to train. For young males the gym is popular to improve physical performance, build muscles and enhance endurance (Upton-Clark, 2024). With this interest, the gym culture and high-intensity training have grown.

While the gym provides an environment where you can work on your physical fitness. There are also a few risks with this interest. The training in the gym can be characterized by the repetitive movement, heavy lifting and unfamiliar exercises. All of these factors add extra strain on the body. When performing the exercises with an improper form, inadequate warm-up increases the risks for injuries. Young males, aged between 18-25, have the highest risks of injuring themselves in the gym (Cuthbertson-Moon et al., 2024). Where the most common cause of the injuries are lifting, carrying, strain on the lower back. Another possible cause for the injuries is the risk taking behavior in the gym and the combination of training in the gym and the day to day physical workload (Noteboom et al., 2023).

A fitting solution would be an app in which a training schedule is made while keeping the day to day life in mind. As well as real time movement correction that is done through the camera to ensure the correct form when performing exercises. Due to a time constraint we focused on building a system that does the real time movement correction. To be able to recognize body parts/joints an AI is used. In addition we set out to train an AI that would be able to differentiate the phases of a deadlift and if a deadlift was correctly performed. However due to a time constraint we decided to calibrate the system ourselves and not use an AI.

To find the value of our solution we first conducted a context analysis where we found that the most injury prone potential users in our research group were the ones that go to the gym on their own. We also found that the acceptance of feedback, in our research group, is based upon the way feedback is given. There is also no app on the market that offers real time movement corrections for the consumer. However in literature real time movement correction is covered with a focus on the feedback acceptance and the accuracy of the softwares (Mukandan, C. et al., 2023). During the low-fidelity testing (LO-FI testing) we found that the feedback mechanism was found valuable by potential users, however the colours were counter intuitive and the feedback of the lines had to be updated more often. In the high fidelity testing (Hi-fi testing) it was tested if a tutorial video with an explanation of the use of the system was needed for users to use the system correctly or if the system was clear enough on its own to use without. It was found that the system needed to be more apparent when a participant had an incorrect posture. Certain visual design choices had to be changed to make the overall use of the system clearer.

The system provides a unique, risk-free, and personalized approach to practicing the deadlift. It allows users to train safely and efficiently while also reducing costs. The system offers real-time feedback where users can immediately improve their posture while doing the exercise.

Chapter 2: Context analysis and concept

As described in the introduction young males aged between 18 and 25 are most likely to get injured when going to the gym. A fitting problem statement within this scope would be: *young males, aged between 18 to 25, are most likely to get injured when going to the gym where the most common cause of injury is lifting/carrying/strain on the lower back*. Revolving around this we formulate a possible concept which would make use of real time feedback and AI. Based on this, we formulate the following research question: *How can a feedback system that uses AI, prevent injury and improve gym performance for young males, aged between 18-25, that are new to going to the gym and tend to go alone?* To get a thorough understanding we broke down the main research question into the following subquestions:

2.1 Research Questions

1. What different information does AI give regarding movement correction in comparison to a sport professional?
2. How do young males, aged between 18-25, deal with feedback on their sporting skills?
3. Do young males aged between 18-25 find it helpful getting sports instruction, to increase physical movement, through a camera?
4. How do adults aged 18-25 perceive the trustworthiness of an AI application designed to monitor and correct movements during gym exercises?

2.2 Approach

To be able to answer the main research question the three sub questions will be answered first. To find answers on the first three sub-research questions a literature study will be conducted. This study will be conducted using literature search engines, Google scholar. For the literature a few guidelines will be established. The publishing date of the literature cannot be earlier than 2010. As well as that a quantitative study needs to have been peer reviewed. For the second research we plan on answering it through literature, however it is possible that is not possible in that case a survey will be conducted. Subquestion four will first be research through literature, but additionally it is planned to have interviews with the people in that age category. After answering subquestion one to four the main research question will be answered through the conducted literature study and interviews.

2.3 Context Analysis: Literature and Related Work

What different information does AI give regarding movement correction in comparison to a sport professional?

To answer this question, we investigated the overall trustworthiness of AI. While we could not find specific research comparing AI's information to that of sports professionals, we focused on AI's reliability and trustworhyness.

What does it take to trust AI? You need evidence that it works in all kinds of different circumstances. This is already implemented in everyday things. Like the food that we buy in the supermarket and the information that is written on the descriptions. In the case of AI, engineers can use mathematical proofs to provide insurance. But this way of working is hard to provide for certain technologies and usages (Caltech, z.d.).
Today's AI cannot think for itself. It is programmed to behave in a certain way. The instructions that the engineer gives the AI system are therefore very important. If the instructions are not good or clear, the AI's learned behavior can have unintended side effects or consequences. Instructions are also used to program values such as fairness into AI models. The people building the model have to choose a definition of fairness. You cannot have a system that is fair in every possible way. It needs to prioritize certain measures of fairness over others in order to output decisions or predictions (Caltech, z.d.).

This research question investigates whether AI provides advice comparable to that of a human professional. Literature suggests that AI does not always deliver accurate results, which is a significant factor in the lack of trust people have in AI (Chakravorti, 2024). Often accuracy gets confused with truth, but high accuracy does not equal truth. This distinction is important, as false results can be dangerous, especially since AI grows in influence every day (Marwala, 2024). While accuracy in AI is a simple and measurable concept that gives the illusion of reliability, it does not guarantee truthfulness. Even if the model's predictions are accurate within the range of its data, it still can be false due to external factors it cannot predict or account for (Marwala, 2024).

How do young males, aged between 18-25, deal with feedback on their sporting skills?

For people to get better at a certain skill or ability, constructive feedback is needed. But how is feedback given in the correct way to young adults and how do they respond to that? For this research question we could not find anything in particular for young males between 18 and 25 years, so we decided to focus on student athletes, as for both target groups feedback is an essential aspect of learning.

To understand a person's reaction to specific feedback, research on motivational climate within the athletic field focused on mastery and ego achievement has been conducted (Smith, Smoll, & Cumming, 2007). According to this study, in a mastery-involving climate, success is defined by self-improvement, mastery, effort and learning. However, ego-involving climates focus on competence and 'natural' ability, where the focus is laid on winning, rather than improving (McArdle & Duda, 2002). These two different perceptions are then responsible for how athletes come to understand their achievement situations and how they will then approach these.

Certain types of motivational feedback have been found to improve athletic performance. According to a study performed by (Mueller and Dweck, 1998), the effects of praise for intelligence and effort were measured on students. It was concluded that students given mastery-involving feedback had more task persistence and enjoyment than students given ego-involving feedback. As well as another study conducted by (Harwood et al., 2015) meta-analysis of 104 achievement motivation theory states the favorability of mastery-involving motivational feedback in specifically sport environments over ego-involving feedback, as those concluded in maladaptive psychological outcomes, as amotivation and anti-social attitudes.

Furthermore, it was made clear in these studies that mastery-involving feedback emphasizes the importance of hard work and effort facilitated improved performance, whereas ego-involving feedback focuses on natural ability. The mastery-involving feedback definitely scored higher than the ego-centered feedback as the first focuses on putting in effort (like training) and the last focuses on the natural ability of the athlete, based on (Moles et al., 2017).

Studies on correcting movement using camera-based body tracking technology revealed that users found the system helpful for receiving corrections. Seeing themselves performing the movement incorrectly, along with visual guidance for improvement, supported them in refining their movements (Conner & Poor, 2016).

How do adults aged 18-25 perceive the trustworthiness of an AI application designed to monitor and correct movements during gym exercises?

While AI (artificial intelligence) technology becomes more common in various domains, so does the need to research the trust in these technologies. While using AI for personalised advice seems promising, it can be complex and raise a lack of trust and skepticism among users. Trust is necessary for human-AI interaction. However, human-AI interactions work differently than human-human Interactions. This is because most people initially assume that AI is (nearly) perfect. Faith forms the essential element of trust and when the number of human-AI interactions grows, faith is replaced by dependability (Sethumadhavan, 2018).
While humans tend to trust AI relatively easily, humans are significantly less tolerant of errors, this leads to bigger declines in trust when a mistake is made (Xie et al., 2019).

An AI system relies heavily on the quality and timing of its feedback to improve its usability. A recent research article suggests that users found that real-time feedback through voice comments or comments after each step-by-step guidance during an exercise could improve trust or compliance (Gadde et al., 2011).

User factors, such as age, culture, gender personality and internal factors also influence trust in AI. A user's mood or self-confidence level could also impact their trust in AI. With higher self-confidence leading to potentially reducing dependence on AI (Sethumadhavan, 2018).
Generations that have grown up with technology tend to feel more comfortable with AI, often leading to higher levels of trust. This comfort level can reduce fear and increase the willingness to follow AI advice, for example an AI application designed to correct movements during gym exercises (Ryan, 2020).
Nevertheless, despite their familiarity, there remains a general lack of awareness and understanding of AI among all age groups, which can influence their trust negatively (Ferrario et al., 2019).
To improve trust in AI, it is important that the system is transparent and explainable. Systems should provide clear explanations for their outputs. This helps users understand how decisions are made and will build trust in the system’s reliability (Toreini et al., 2020).

This research question is intended to find out how adults 18 to 25 perceive the trustworthiness of an AI application designed to monitor and correct movements in the gym. However, there was no research found for this specific age group. This is why we relied on broader research and thus a broader age group. While this research provides valuable insights, the conclusion should be interpreted with the knowledge that it does not just apply to the people aged 18-25 but to a wider range. In conclusion, adults perceive the trustworthiness of an AI application designed to monitor and correct gym exercises based on a combination of elements. The AI's accuracy, the timeliness and clarity of feedback, the user's characteristics, and the system's transparency and explainability all influence the user's trust.

Do young males aged between 18-25 find it helpful getting sports instruction, to increase physical movement, through a camera?

This research question was answered through interviews since we were not able to find corresponding literature. Based on the answers of the interviewee's we can conclude that while males aged between 18-25 are open to receiving sports related feedback, they are skeptical about AI-based feedback or advice, see chapter 2.4. The static of the camera and its limited angle of sight caused them to be hesitant towards the feedback, see chapter 2.4. The interviewees said they preferred constructive feedback and feel that AI might be more suitable for personalized work-out plans or nutrition advice, see chapter 2.4.

2.4 Context Analysis: Interviews and Observations

To be able to understand if the findings in literature were the same as in real life we conducted multiple interviews. In addition, we also set out to understand if our target group needed to be more specified.

For our interviews we came up with a semi-structured interview method. By using a semi-structured approach, the interviews were given a certain lead by the interviewer, as well as the interviewee. The participants were asked all the prepared questions, as well as being able to add information towards the interview for more clarification, see appendix 3: interview questions, for the pre-phrased interview questions, as they were asked during the interview. For the interview we interviewed four male participants, aged between 18 and 25 that often go to the gym. During the interview, clear and open questions were asked, following a script. We chose to follow a pre-phrased script to cover all important questions and themes. Two of the questions we asked were based on a sub research question, because we were not able to find literature to be able to answer the question. To clarify a participant's answer we asked additional questions to fully understand our research and gain important key events/takeaways. After conducting all the interviews, we carefully summarised the main takeaways into four summaries.

When going over the interviews we observed that even though all of our participants were male and aged between 18 and 25, certain differences in the outcome could be found, which could have been encountered due to demographic differences of participants. Participant one and two often go to the gym together with other people and never experienced bad injuries before. Participant three on the other hand mostly works out on his own and experienced injuries in the back and shoulders, due to improper posture, see appendix 4: summaries of conducted interviews. Another noteworthy finding was that all participants do not mind getting feedback if the feedback is meant for improvement rather than for nagging or as an egocentric means. Additionally, participants do not like to get detailed feedback on exercises they are familiar with, as this does not encourage them to grow, but rather annoy them instead. Another finding is that no participant was using heavier weights for ego-lifting, as all of them are experienced sporters that know how important it is to keep their own training schedule going rather than impress other people. Participant two even mentioned that he would only use heavier weights with “training buddies”, as they can “spot” him and so guarantee more safety for the exercise. When participants were asked if they found it helpful to get feedback through camera, based on sub research question three, participants were hesitant due to the staticness of the camera and the limited angle of sight of the posture. The participants had a positive mindset; however, they did not fully believe that the AI could faultlessly improve their form during working out. Instead, they came up with ideas where the AI would be a better suggestion. It could predict the perfect food plan for people going to the gym, or a workout schedule.

The key takeaways from the interviews conducted are that we decided to focus more on people going to the gym on their own or using a house gym. Due to the fact that participants who do not do sports on their own in the gym already often receive feedback from others. We also found that the way feedback is provided is important and can influence the acceptance of the feedback by the user. The participant noted that uplifting- and specific tones are preferred.

2.5 Description of Target User Group and Domain

Males aged between 18-25 are the most represented demographic in gyms (Cuthbertson-Moon et al., 2024). This group is likely to engage in strength training exercises, particularly weightlifting. Weightlifting is strongly associated with injury, especially lower back injuries (Cuthbertson-Moon et al., 2024). Young males are also the demographic most likely to take risks in their workout routines, such as skipping warm-ups or lifting heavier weights than they can manage without proper form (McCallum, 2022). There exists a phenomenon called ‘Ego-lifting’, which occurs when individuals prioritize lifting heavier weights instead of the correct form, which increases the risk of sustaining injuries.
Furthermore, fitness influencers often recommend going “all out” and ‘training until failure’ during a workout to maximise muscle growth. This method encourages individuals to continue doing repetitions until they are no longer able to perform a repetition with proper form. This ‘training until failure’ can come at great cost, this can be extremely straining on the body and can cause overtraining of certain muscles leading to a higher risk of injuries (Bobo, 2024).

Fitness apps have become a popular medium to track activities and have almost become essential, especially among young adults. Popular fitness apps among young adults are apps like Nike+ Training Club, Fitbit and MyFitnessPal. Fitness apps like these can track things like workout guidance, calorie intake and heart rate tracking. These apps are not only able to improve the regularity of working out, but long-term use of certain fitness apps will also improve their user’s physical, emotional, social and cognitive status and promote their well-being (Cai & Li, 2024). With a growth in revenue of 0.68 billion dollars in 2016 to 5.35 billion dollars in 2021, the fitness apps have become one of the most popular applications the app store has to offer. Additionally, in 2019, the average person spent around 2 hours and 15 minutes per week using fitness apps, which is a significant growth compared to the average of 1 hour and 45 minutes in 2018 (Muscle, 2024). While there exists a wide range of fitness apps in the app store, most of the apps are focused on exercise tracking instead of injury prevention. This is a significant oversight as injury prevention is an important aspect of exercising and achieving fitness goals safely.

While Fitness apps are currently very popular among males aged between 18-25, there are other popular mediums. Smart gym equipment is getting more popular for this demographic as well. Smart gym equipment can have capabilities that fitness apps do not have, they may have the potential to enhance the athletes' performance or reduce the risk of injuries. However, most installations in the gym are focused on tracking things like range of motion (ROM), the number of repetitions or the intensity of the work-out. This means that there is limited attention to injury prevention.
Given that the lower back injuries from working out in the gym are highly associated with improper form, there is a need for a solution that gives real time corrective feedback on form while deadlifting. Recent advancements in sport related technologies are things as computer vision and wearable sensors, these have demonstrated the potential for accurate posture assessment in real-time settings (Bagga & Yang, 2024)

Due to the constraints of time and resources, we focused on a real time feedback system rather than an app. This system allows for a more focused and immediate application of form correction, that requires minimal user input, while also addressing the need for injury prevention in the gym. Our decision aligns with the preferences of young male gym-goers, as they prioritize convenience and immediate results (McKane, 2024).

Scenario 1
A 23-year-old man is following the 'training until failure' method continuously, ignoring his physical limits. He develops a lower back injury due to the overstraining of the back and improper load management.

Scenario 2
An 18-year-old gym-goer goes to the gym with his friends. To impress them, he attempts to deadlift a heavier weight than he is capable of. The 18-year-old compromises on form and strains his lower back.

Scenario 3
A 25-year-old man goes to the gym before work regularly which means that he does not have a lot of time. This is why he compromises on warming up and cooling down, neglecting his body's needs and experiencing discomfort during his exercises due to tight muscles.

2.6 Product Concept and Global Requirements

Prior to finding the product concept and solution, an ideation process was held. Throughout the ideation we came up with a user domain and a technology domain, giving us a five-by-five matrix with possible combinations of technologies and users. As part of the user domain we selected 'students', elderly', '30+ working', 'tourists' and 'pro-athletes'. The Technology domain included, 'App', 'Robot', 'AI', 'Sensors', 'Audio & Visual Technology', see appendix 1: 5 by 5 matrix.

After looking at all the possible user domain and technology domain combinations in, see Appendix 1: 5 by 5 matrix, we decided to select students as a user domain with the app. Our decision was based on the fact that students have a big demographic, and that movement is very important for them, as students tend to sit more and more nowadays. We decided to combine this with an app, as apps are budget friendly, easily accessible and already have been tested for movement correction through cameras. As we tried to approach the concept further, we took a step back and found that not all students are regular gym-goers, as additionally the most injury prone group in the gym are young males aged from 18 to 25 (Cuthbertson-Moon et al., 2024). They tend to challenge each other and conduct risk taking behavior (Noteboom et al., 2023). This way we changed the user domain to young males, aged between 18-25 that regularly attend the gym. Based on the chosen user group we thought of 50 ideas to solve the problem statement: “Young males, aged between 18 to 25, are most likely to get injured when going to the gym, where the most common cause of injury is lifting/carrying/starin on the lower back (Williamson, 2024).

After looking at all our ideas (Appendix 2: 50 ideas), we decided to do a mix of multiple ideas. Our first concept of the solution is an app for males between 18-25 years that regularly visit the gym. With this app we intend to have multiple functions that reduce the risk of injuries while attending the gym frequently. Some incorporated ideas include AI to create customized training schedules for individual users that keeps daily life into account. Also sensors of the phone will be included, while correcting the correctness of body movement through the phone sensors and a camera. The app is used as the main technology, as it can be easily combined with earlier mentioned technologies, as an additional function. Some global requirements set for the app are minimalistic and clear design, that is easy to use for everybody. Our user group is either working or a student and neither have a lot of time to learn how the app works, so it should be very straightforward.
To have a benchmark of the minimal requirements of our system we have formed a list of user requirements, system requirements and constraints. These requirements and the constraints are based upon the interviews conducted to answer the design questions and the literature research.

User requirements

- With the fitness application young males shall be less prone to gym injuries
- With the fitness application young males shall be able to achieve better form when weightlifting
- With the fitness application young males will be able to see their form and how to correct it while weightlifting in real time
- With the fitness application young males will receive customized training schedules to make staying fit easier for them

System requirements

- The system must know the correct position of certain weightlifting exercises
- The system must have a simple design that is easy to understand for new users
- The system must be able to compare the user’s position to the correct form of an exercise
- The system must be able to locate the user’s position of each limb
- The system should be able to show the position of the user on the application

Constraints

- The footage must be deleted shortly after use to address potential privacy concerns
- AI will not use or store the user's data and footage

Chapter 3: Design and Evaluation of Lo-fi Prototype(s) and Revised Product Concept

After meeting with the project mentors and thoroughly reviewing the concept, we have decided to move forward with a gym installation concept. We considered the installation to be the most interactive solution and the most realistic to bring to life. Time constraints played a significant role in our decision-making process. Given the limited timeline for this project, the installation allows us to focus on delivering a functional and effective product.

The first concept of the gym installation is focused on improving the deadlift form in real time of a gym-goer. There will be a camera filming the user from the side, the user will be able to see their form on a screen in front of them, on the ground, as this is how the deadlift is best performed. With lines going through their body and joints, as well as other lines drawn to indicate the ideal deadlift form, so the user can alter their position based on these lines. Red lines will indicate the current position, and green lines will indicate the ideal position.

3.1 Lo-Fi Prototypes

Our goal with the low-fidelity (lo-fi) prototype is to evaluate whether users understand the primary visual feedback presented on the screen, see figure 1. Specifically, we aim to determine if they comprehend the lines displayed to correct their form during a deadlift. The research questions provide essential context and further insight into the topic, focusing on how young males perceive feedback.

Our target audience for testing the prototype consists of young adults aged between 18–25. This initial test is solely intended to assess whether the visual feedback is effectively understood and if users can interpret the meaning of the lines on the screen. We conducted the evaluation by presenting our video prototype and then asking participants a series of questions to understand how they perceived the feedback.

To summarize the results from the lo-fi prototype testing are that the participants were confused, the first time they viewed the video. After viewing the video multiple times and prompting them about the visual feedback they understood what they were looking at. A big difference in understanding the visual feedback was the prior knowledge of the deadlift. A reoccurring point of feedback was the colour choice of the line that was following you. During the prototype testing it was red which the participants instantly linked to doing something wrong. In addition, a participant suggested that the correct form could be updated more often during the work out, see appendix 5: interview answers lo-fi prototyping.

To conclude, the form of visual feedback that we tested during the lo-fi prototyping was understood. However, the visual feedback needed a bit more context, for example instructions, to be understood easier by participants. In addition, the colours used to display the user's movement, and the correction of the movement must be considered carefully. Lastly the feedback for a good posture should be updated in more frames, so that the user knows which form they should have during different moments of the deadlift. Participants noted a delay in the positioning of the lines. However, this issue will not occur in the final product, as it will not rely on screenshots for different frames. Additionally, this is a straightforward issue to address in future prototypes.

In future prototypes, we plan to experiment with different colours to observe how various hues influence user reactions. Additionally, we aim to test different camera angles to determine which perspective provides the clearest and most effective visual explanation.

Lo-fi Prototype

Figure 11: in this figure three screenshots are displayed of video prototyping used during the lo-fi. The three screenshots displays the three different movement phases with the visual correction.

3.2 Revised Product Concept and Requirements

After reviewing the results of the lo-fi prototype evaluations, the product concept was adapted. From the interviews it was clear that the visual feedback had to be explained beforehand, and therefore the user will be asked a few questions before using the installation. First, the user will be asked whether they know how to perform deadlifts. If not, a short video will appear explaining how to do the deadlift from both the front and side view, how far apart the user's feet should be and the importance of staying upright. Therefore a front mirror is needed and the side profile will be shown on the screen in front of the user, placed on the floor to optimize the deadlift form. This way, the deadlift position is corrected from the side through software and filmed with a (phone) camera. The user is then able to correct its position in real time with the front mirror (display on the ground). The installation should be placed in a corner of the gym to ensure the camera or software does not confuse the user with a bystander, but the ideal placement has yet to be confirmed through evaluations. However, we aim to program the software in a way that it can always distinguish this difference.

The colours going through the user's body and joints on the lo-fi prototype were green and red. This confused the interviewees during the prototype evaluations. As a result, the red colour was changed to blue, indicating the user's current position. The green line shows the optimal deadlift form and will disappear when the user gets (a certain range) close to this form. These lines will move simultaneously with the user's position and will be shown on the screen in real time. The user will also be asked if it is their first time using the installation. In that case, a short introduction video will be shown explaining the visual feedback.

An additional feature could be adding lights around the screen in front of the user, that will turn on when the user could be improving their form, turning their attention to the screen. This feature was tested in the hi-fi prototype.

The footage will not be stored, so there is no uncertainty for the user and what might happen to their data. New global requirements can be formed from this altered concept. Listed below are the new user and system requirements and constraints.

User requirements

- With the installation the young males who are new to the gym and usually train alone shall be guided step-by-step in understanding how to use the system before starting their workout
- With the installation the young males who are new to the gym and usually train alone shall be less prone to lower back injuries
- With the installation the young males who are new to the gym and usually train alone shall be able to achieve better form when performing deadlifts
- With the installation the young males who are new to the gym and usually train alone will be able to see their form and how to correct it in real time while deadlifting

System requirements

- The system must know the correct deadlift position
- The system must ask the user whether they are familiar with deadlifting and if the user is not, explain how to perform this exercise properly
- The system must be able to compare the user's position to the correct deadlift position
- The system must be able to locate the user's position of each limb and calculate the angles between
- The system must be able to send the data/footage from the side camera to the calculation program and then be able to send this corrected footage to the screen in front of the user in real time
- The system should be able to show the position of the user in real-time on a screen
- The system should be able to get the user's attention to the screen with lights so the user knows their posture can be improved
- The system must be able to distinguish the user from other people and gym equipment in the camera's frame

Constraints

- The system must comply with relevant privacy laws and regulations, ensuring user consent and secure processing of data.
- The screen in front of the user should be placed on a safe distance so it cannot break

Chapter 4: Design of Final Hi-Fi Prototype

Goal of hi-fi prototype

The primary goal for our high fidelity prototype is to test whether the users of our installation need an instruction video to interact with our system successfully (which can be defined in understanding of the task, the received feedback of the system and the goal of the installation). By comparing two groups, a group with an instruction video and a group without, we aim to determine the comprehension of our user groups and the relevance of showing the instruction video.

Explanation of Prototype on higher level

The hi-fi Prototype (Live Lift) aims to reduce the risks of lower back injuries during deadlift exercises for young males between 18 and 25 years. As the user pursues a deadlift, the posture is being recorded from the side (profile) with a camera, while the user then sees its profile displayed in front of them. Coloured lines then follow the movement of the user's body on the screen and calculate the difference between the user's limbs positions based on predefined ideal measurements in different phase(no deadlift, beginning 1, beginning 2, mid deadlift, end 1, end 2) of the deadlift movement. An example of this can be seen in figure 2 As the user's posture deviates from the predefined ideal for a deadlift, the blue line (line of the user) colours red (wrong posture) and a green line appears (ideal line). As the user improves its posture towards the ideal measurements, the red line turns blue again and the green line disappears. As long as improvements can be made, LEDs light up on the screen in front of the user to draw attention towards it. This way the posture can be improved in real time and risks of injuries will be reduced, as users are able to correct themselves as they do the workout.

Figure 2: This figure displays the screen used during prototyping and the LED encasing.

In figure 3 the interaction of the system with the user can be seen in an architecture diagram.
As the user starts doing the exercise, the camera feed goes through the openCV(video stream) and the video gets processed. When the camera stream comes in, the user's key joints (left toe, left heel, left ankle, left knee, left hip, left shoulder and left ear) get defined in the video footage through a library called mediapipe. As the joints are found, the form checker logic applies calculations to determine the movement phase in which the user is, based on predefined correct key points and compares them to the extracted key point from the video footage. Then the 'good form function' compares the angles of the limbs extracted from the video feed to the predefined angles corresponding to a good form when executing a deadlift. In the meantime, the system calculates the length of the users limbs and the optimal position per limb based upon the predefined angles of a good form. If the form of the users, when executing the movement, is correct then the user only sees the blue line. When the form is incorrect the users see lines showing them corrections based on their posture. Simultaneously the LEDs work as feedback to the user, turning on whenever the ideal posture is not met and attracting their attention to the screen.

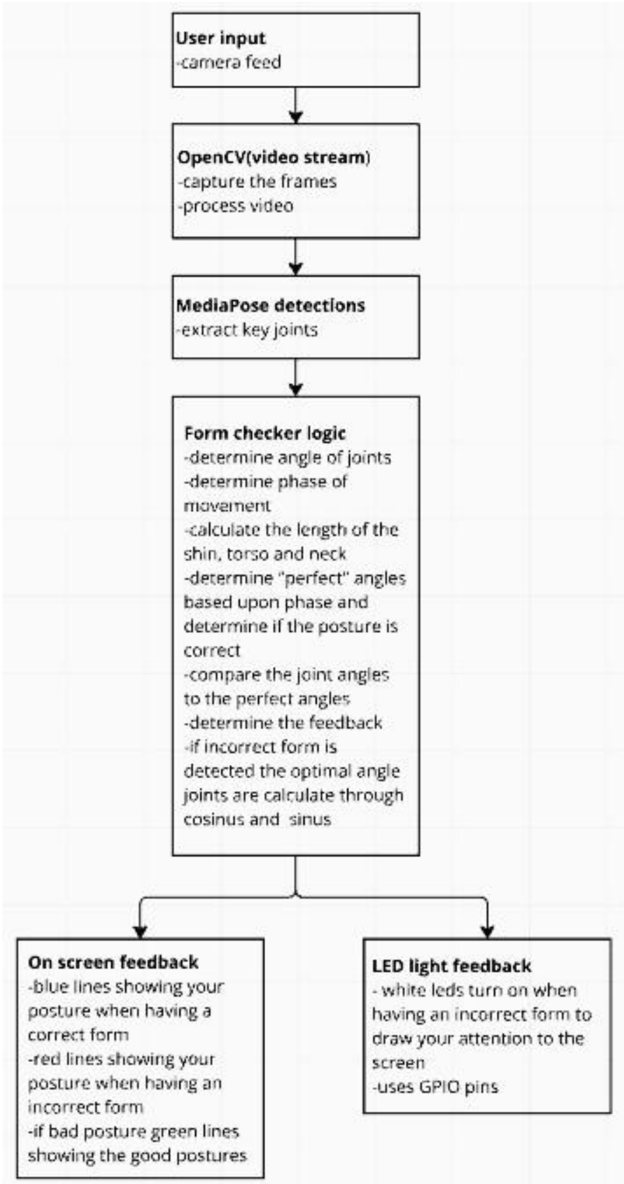

Figure 3: In this figure the system architecture is displayed.

The set-up of the Hi-Fi prototype includes the following:

- A metal bar: To prevent injuries we opted for a lightweight bar. This offers a safer alternative for the standard gym bars, which typically weigh 15-20 kilograms, see figure 7.
- Two stools : These stools are placed to support the metal bar to position it at an appropriate height to perform a deadlift and prevent the participant from overstraining their back, see figure 6.
- A tripod with a mobile phone camera: The tripod is positioned around 2.5 meters away from the metal bar. The camera is used to record and send the participant's movements to the laptop see figure 5.
- Laptop with LED encasing: This laptop is placed on the ground near the set-up and is encased with a frame with white LED lights, see figure 4.
- Python script: A python script is run on the laptop and lets the laptop provide visual cues and feedback during the deadlift, see appendix 10: deadlift corrector script.
- Arduino: This arduino is connected to the laptop and contains the script with logic to turn the LEDs on and off, see appendix 11: arduino script .
- Encasing: Designed on Fusion360, see appendix 12 for the .f3d file, and laser-cut. 11 LEDs soldered in parallel were built in afterward.

Figure 4: In this figure the screen with LEDs as visual feedback.

5: This figures shows the setup of the camera

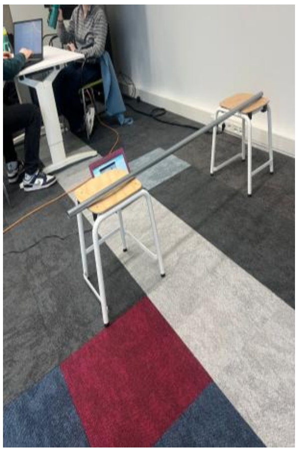

Figure 6: This figures show the set up of the barstools and the bar.

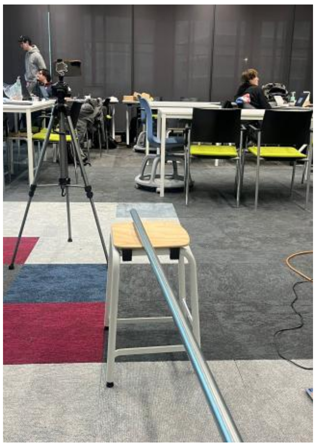

Figure 7: The figure show the set up from the right side.

As discussed in chapter 3, during the low fidelity prototype test, the colours going through the body on the screen were red and green. However, these colours confused the user, as they already have a predefined meaning. Red and green are used in everyday contexts, such as user interfaces (error and success) or traffic lights (stop and go). Also, red is associated with failure or danger which could have let users to feel more discouraged, (Hong et al., 2020). To solve this issue, and improve the usability, we changed the colours to blue and green. In contradiction with red, blue is often associated with openness, peace, and tranquility. Blue is more likely to activate motivation, because these associations with blue signal an environment that encourages people to use innovation as opposed to a more tried-and-true mentality (Mehta & Zhu, 2009). Green remains the colour of the ultimate form, as it is usually associated with correctness and success (Moller et al., 2009).

We applied usability heuristics to improve the usability of our prototype. For this we reference Nielsen's 10 Usability heuristics principles. Which have been proven to increase a system's usability (Mankoff et al., 2003). We have applied the following heuristics in our design process of the high fidelity prototype:

1. **Visibility of system status**
   This heuristic refers to the importance that the user should always be informed about the underlying process to what is happening, as they have performed an action (Nielsen, 2024).
   For our high fidelity prototype we have focused on this principle by adding LEDs above the display, where users get instant visual feedback of the LEDs turning on when the posture is incorrect. This way there is no need for them to stop moving and analyse complex instructions and users can improve their posture in real time because of the visibility of the system status.

2. **Flexibility and efficiency of use**
   Ideally the system should accommodate both beginners and experienced users, this ensures that every user can use the installation efficiently and effectively (Nielsen, 2024).
   During our low fidelity testing we have encountered people with no experience of the deadlift got more confused with understanding the goal and function of the installation than more experienced people who knew how a deadlift is performed. To ensure that our system works for users with no prior knowledge as well, we included a QR code (Quick response code) that links towards a short instruction video that explains what a deadlift is and how it is performed. This gives users a choice on whether they want to scan the qr code and watch the instruction video or start with the interaction of the system right away.

3. **Aesthetic and minimalist design**
   No complex and unnecessary elements should be included, to reduce distractions and cognitive overload for the users(Nielsen, 2024). Therefore, the interface of our prototype focuses solely on the tracking of the movement, without cluttering the screen with extra information that could possibly distract the user. We also used a limited colour scheme to avoid any confusion and we kept the LED feedback simple with only white coloured lights.

4. **Recognition rather than recall**
   Users should not have to remember too much information on specifics, on how to use and interact with a product/system (Nielsen, 2024).
   By choosing colours like green and red, correct and incorrect, for the feedback lines on the display that users interact with, they are able to recognize the meaning behind them easier. Also blue usually has a natural meaning, nudging the user into the right direction.

5. **User control and freedom**
   Users should feel in control of their actions and be able to adjust their experience according to their needs and wishes (Nielsen, 2024).

Our prototype does not force interactions. Users can step up, do their exercise and leave whenever they want to. Users of all prior knowledge levels can choose whether they want to see the instruction video on how to do a deadlift or not. It is also possible to use this installation multiple times to get a correct posture. After some repetition the user already knows how to use the installation and does not require to see the short instruction video about the meaning behind the feedback received on the display anymore. In this case the user can solely start with the workout directly. This makes the experience more self-directed than restrictive.

We chose an in-between-subjects test for the Hi-fi prototype because this method ensures that each user is exposed to only one condition, either with or without the instruction video. This is to avoid learning effects, where the participant could be able to get familiar with the system, and therefore make it difficult to determine if the instruction video itself had any influence on the understanding level of the received feedback and goal of the installation.

Qualitative research was used to obtain more in-depth insight into the usability of our installation, rather than just numerical data. Since the high-fidelity prototype involves visual feedback, and real-time interaction, it is important to understand how the user views the system. Hence, in addition to testing the intuitiveness of the system and the importance of the video, we also added questions about the LEDs and overall performance questions to the post deadlift form (appendix 7 and 8).

Chapter 5: Evaluation - Experimental Research Study

5.1 Evaluation plans

The evaluation was done simultaneously with the Hi-Fi prototyping. The goal and research questions have been evaluated in previous chapters.

Goal of research experiment
The goal of the Hi-Fi prototype testing was to determine whether an instruction video explaining how the system works would enhance the user experience or if it could be omitted. Additionally, we aimed to gather feedback on specific design choices and assess whether participants understood the meaning behind certain actions performed by the system. For example, we wanted to know if users recognized that the LED indicator turning on signified an incorrect form.

Hypothesis
The hypothesis is that participants do need the video to use the system smoothly. However, we also believe that it is not entirely necessary to see the video to understand what is going on visually. Design elements in the system should provide a certain idea about what is happening.

Experimental study design, including setup
For the Hi-Fi test we conducted an in-between test. In total we tested the prototype with eight participants, all males and between the ages of 18-25. The test was done in two groups, thus using the in-between method, so that there would be no learning effect. The effect of the instruction could be better studied this way. The first group did not see the instruction video of our prototype and was asked to do the exercise without any extra information about the system. After testing the system, they were asked to fill in a form to gather information about their experience with the system. The second group was shown an instruction video of the system and what was expected of them. Each participant was asked to fill out a questionnaire. The questions were the same for everyone, except for those who watched the instructional video, as they were also asked how well they understood the installation and how clear the video was. All the participants were shown a video about what a deadlift is and how to do it so everyone's knowledge about the exercise was on the same level.

The set-up of the hi-fi prototype has been described in depth in chapter 4, but an overall description includes the following:

- A metal bar: To prevent injuries we opted for a lightweight bar. This offers a safer alternative for the standard gym bars, which typically weigh 15-20 kilograms, see figure 6.
- Two stools: These stools are placed to support the metal bar to position it at an appropriate height to perform a deadlift and prevent the participant from overstraining their back, see figure 6.

- A tripod with a mobile phone camera: The tripod is positioned around 2.5 meters away from the metal bar. The camera is used to record and send the participant's movements to the laptop, see figure 5.
- Iriun app: This app is installed on both laptop and phone and is used to make the phone a remote webcam.
- Laptop with LED encasing: This laptop is placed on the ground near the set-up and is encased with a frame with white LED light. This laptop provides visual cues and feedback during the deadlift, see figure 4.
- Arduino: On the arduino a script is uploaded that contains the logic to turn the LEDs on and off, see appendix 11: arduino script.
- Python script: On the laptop a python script is used that determines if the user has a correct form while executing the deadlift, see appendix 10: a deadlift form corrector.

Interaction protocol
For our hi-fi prototype testing we performed an in-between subject test. We divided the groups into two groups of 4: *Group A, and Group B*.
Both groups are given a consent form first thing, see appendix 9: consent form Hi-fi user testing.
As the user enters the installation (hi-fi prototype), we will ask them whether they are familiar with the deadlift. If the question is denied, a deadlift tutorial video, see appendix 13, of it is shown, explaining how the correct posture is achieved.

Group A
The participants of *Group A* are not shown an instruction video of our installation. They are also not provided with any other information about the installation while doing the exercise or before that.

Group B
Group B is shown an instruction video of the installation, the link may be found in appendix 13, after filling in the consent form. In the instruction video, the visual feedback is shown for first time users, to eliminate all possible misunderstandings beforehand (wrong interpretation of lines, colours, correct positioning at the installation). When the user has watched the video(s), they are asked to move towards the installation.

Group A and B
The user is then supposed to take the metal bar and perform six to eight deadlifts according to their knowledge or the instruction video. A phone camera on a tripod from the side will record the user's posture and movements and compare it to the ideal posture, based upon an ideal scenario. A screen is placed in front of the participants to show their form and the visual feedback. As the user moves, the lines should stay blue, as this indicates that the ideal posture and the participant's posture match. If the blue lines turn red and green lines appear, white LEDs light up, indicating attention to the screen and that the user should re-adjust its posture towards the green line until there is only one blue line, indicating the perfect deadlift.

For *Group A* (no video), their performance predominately measures the intuitiveness of our prototype system.

For *Group B* (with video), their performance and interaction with the system reflects how much the video enhances their understanding of the system.

Both groups were asked to fill in a form (appendix 7 and 8), with questions about their experience with the system. The main topic of this hi-fi prototype was to test if the explanation video of the system was necessary or that the participant directly knew what the system was supposed to do. We also asked about some sub-topics. These were mainly involved around design choices. In both forms we asked about the choice of line colours, brightness of the LEDs and overall, what their experience was with the product. For the participants that watched the instruction video about the system we added some additional questions:

- What did and did you not like about the instruction video?
- What do you think of the voice speaking in the video?
- Was the instruction video clear?
- Was it clear how to use the installation after watching the video? Explain your answer

Constructs

The form is the main method that we used to evaluate our system. Here the participant could state their opinion of the system and we had a clear informative answer that we could later look back on to evaluate the system's usability. However, we also looked at the participants while they were doing the exercise and how they reacted to the system. This was especially valuable for group A. Their body language was an indication of how well they understood the system's intentions.

Analysis plan

After all the answers from the participants are collected we will compare the different answers from group A and B regarding the clarity of the explanation video. We also analysed the answers that were given by the participants in the forms regarding the design of the product. We mainly focused on clarity and the experience of the user experience.

5.2 User Study

Pilot study

For the pilot study, we aimed to test the setup of our hi-fi prototype testing. The primary goal was to ensure a smooth prototyping process, allowing participants to focus on performing the exercise and reflecting on their experience with the system. This was tested by each of us performing deadlifts and testing the installation, its clarity and effectiveness.

After conducting the pilot study, we concluded that the bar should not be placed on the ground, as this does not reflect the starting position of a deadlift and makes it difficult for the software to grasp this position. Instead, we decided to place the bar on stools for easier access. Additionally, we adjusted the camera angle to give participants a clearer view of themselves performing the deadlift. We also changed the size of the window on the screen placed in front of the user, as we found that it would be easier to see the visual feedback if it was bigger.

Results

Overall, the people that watched the instruction video (group B) had a better idea of what the system was for and how it worked. The people that did not watch the video (group A) asked more often what they had to do during the experiment and their body language indicated that they were confused about the system's intentions while first using it. This was also reflected in the answers that they filled in the forms, for example for the question: How did you feel about the installation? Answers for this question were: "I was a little bit confused about the feedback" and "Very cool, but I don't know what it actually does."

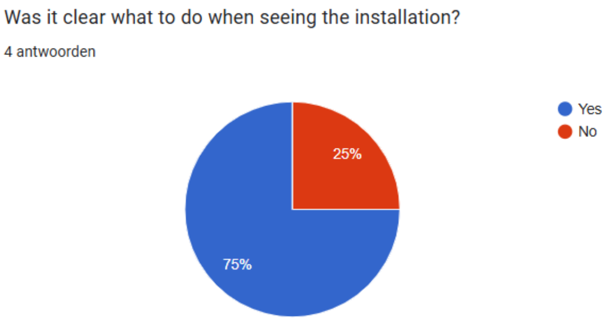

Figure 8: This figure shows the answers from the quesationair question (group A): was it clear what to do when seeing the installation.

There were also questions asked about the design of the system. The conclusion that could be drawn about this was that the LEDs were too bright. Some participants also suggested that we should change the colour of the LEDs, "maybe make them red to show that it's bad rather than just white".

The feedback on the camera angle and how well the participants could see themselves was positive, see figure 9, with answers such as: "It was good, I could see my whole body on screen". Although we also got some comments that people would like to review the feedback for further analysis, "alright. if software/work effort allows it, perhaps consider rearview". There was some confusion in what the lines meant. This was especially visible with group A, "green means good i think" and "a little bit confused". Group B knew what the colour of the lines meant because this was explained in the video but there were comments that suggested changing the colour for more clarity, "I would assume my own line would change red, when I did something wrong." and "I liked that the correct position was shown in green, but maybe make the lines green if you are doing it correctly or red if you aren't".

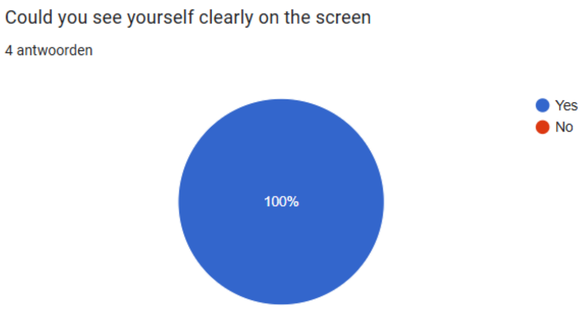

Figure 9: In this figure the answers from the questionair question (group B) could you see yourself clearly on the screen is visable.

The last question that we asked was whether the participants thought this exercise would help them in better performing the deadlift. In both group A and group B we got a positive response with answers such as: "it helped me realize I have to keep a straight back and lean less forward" (group B), "Get a better deadlift posture" (group A). Some of the participants indicated during the experiment that this system could be useful for beginners and that they would not particularly seek a device like this since they already know how to do a deadlift.

Interpretation of the data

Out of the data that we collected with the hi-fi prototype testing, we can conclude that, for clarity, it is best to show the instruction video first before using the system. Although we saw

that during testing the installation, participants eventually knew what to do, but it was better for the experience of the user to know exactly what to do from the beginning.
It was also clear that there would be some design changes necessary for better usability. The brightness and colour of the LEDs would be changed for a better understanding and less distraction. The colour of the lines should also be changed for more clarity about their meaning. The lines are red when the participants are not lifting ideally, green indicates the correct form and blue only when the user does the deadlift correctly.
The instruction video that was shown to group B was clear and helpful and no additional changes would have to be made. The same goes for the angle of the camera and the screen that was placed in front of the user.

Overall, showing the instruction video helped clarify the system’s intention for participants. However, those who did not watch the video eventually understood the system's purpose as well. Our assumption that we did not need the video is therefore correct but it is clear from the results that the participants who watched the instruction video first had a clearer understanding of what the system is about.

Chapter 6: Discussion

During this project we developed a prototype for an installation that could be placed in a gym. The system gives the user real time feedback when performing a deadlift so that they can improve their form. This solution is based on the problem statement *young males, aged between 18 to 25, are most likely to get injured when going to the gym where the most common cause of injury is lifting/carrying/strain on the lower back*(). Based upon this problem statement a context analysis was conducted, with one research question and sub research question. Then a lo-fi testing was conducted with a lo-fi prototype to understand how the design ideas we had affected the usability of the system. After which we have done a hi-fi prototyping test simultaneous with the evaluation.

Our evaluation primarily focused on usability, as we wanted to assess whether people enjoyed using the product, given that its use is entirely voluntary. Conducting an intermediate test allowed us to gather valuable insights from our hi-fi prototype, making the testing process more effective and helping us refine our system.

The hi-fi testing went smoothly, and overall, participants responded positively to the product. We received specific feedback that enabled us to optimize the system. Based on this feedback, we adjusted the LED colour and brightness, as well as the line colours, to improve clarity.

While we cannot definitively claim that our system is a major success—since that was not the focus of our research, our findings suggest that it has strong potential. To validate its effectiveness, further testing is needed, particularly on functionality and efficiency. For instance, quantitative tests could assess whether the system helps reduce pain. Additionally, some aspects of the system remain underdeveloped due to time constraints and the limitations of available equipment.

The uniqueness of our system lies in that the user gets immediate feedback while doing the exercise. The current market competitors to give feedback on the user's form while performing a deadlift are either personal trainers or the mirror in the gym that they use while exercising. Personal trainers are quite costly and with our young age group who are often on a tight budget a more affordable solution would be more attractive. The mirror in the gym is in a fixed position and due to the form of the deadlift where the neck and back should be straight. Thus, the user can only use the mirror when performing the deadlift with the correct form while looking forward. The system uses the camera which is positioned to the side of the user and streams to a screen strategically placed onto the ground to nudge the user to keep their neck and back straight. Thus, the system provides users with a new perspective that allows users to correct their form in real-time.
During the hi-fi test participants came back to tell us how much they like the system and some participants even wanted to try the system again.

When designing this system we had a few limitations, the biggest being the time constraint. In the span of ten weeks, we needed to come up with an idea, do a context analysis, run a lo-fi and hi-fi test and lastly do an evaluation. This means that we had to almost rush through all of the different phases. This also meant that we could only take one piece of our original idea and work that out and had to let the rest go. The time limitation affected us the most during the testing phase. We would have liked to run an additional lo-fi evaluation to further improve the

design choice. This would have made it possible for us to do more quantitative hi-fi testing and evaluation. When doing the contextual research, a limitation was the novelty surrounding AI and interactions with our specific target group. We were not able to answer two of our research questions, due to the limited research articles. The questions could in one case be answered on a nonspecific level and in the other case we tried to answer the question through interviews with our target group. However, there was a limited amount of research that was relevant to use in our case, since it is quite novel.
A limitation when testing was the testing scale during our lo-fi and hi-fi testing. During these tests we mostly used classmates that were male and between 18 and 25. However there is a bias in the group we tested, they have already an affinity for technology and like innovation. This could result in them understanding designs more easily and being more likely to be more open to the concept.
When building the hi-fi prototype for the system we were limited by our technical skills, especially the programming skills. The interaction that we designed was built, however the system is quite slow due to the number of calculations and the low processing power of the computer. The system would perform better if it was AI trained but because of technical skill and shortage of time this was not possible. The system is also very sensitive meaning that it is not performing as it should when multiple people are in the frame or the camera detects some noise.
Limitations of the product would be the delay that the system has between the user's movement and the feedback. This could lead to the users incorrectly correcting their form. Another limitation of the system is the camera that is used and the condition the camera is placed in. The incorrect lighting or not having a good quality camera can impact the feedback given by the system.

A next step to further improve our system would be decreasing the delay between the movement of the user and the feedback given by the system. Currently the delay when moving quickly through the phase of a deadlift the system cannot give accurate feedback to the user. Another improvement would be the camera setting that is used. Currently the camera is placed without thinking of the optimal lighting. When taking this into account the system's accuracy could be improved.

During the hi-fi testing, we used a white background to ensure that the system would function effectively. For future testing, it is essential to evaluate the system's performance with various backgrounds, including dynamic ones, and enhance its ability to operate reliably even in noisy environments.

A future test could be evaluating if the feedback system is able to decrease the number of injuries when executing the deadlift. This is important to be able to make the claim that the system works and is able to decrease the chances of getting an injury. Additionally, it is important to do an evaluation where the acceptance of the system is the main research question of the evaluation to understand if there is an actual market for this product.

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Appendix 1: 5 x 5 matrix

User Domain Tech Domain	Students	Elderly	30+ working	Tourists	Pro-athletes
App	App giving challenges to students to find a specific spot on campus and do some sport exercises like squats against a wall → you can earn points to get a nice price at the spar.	Elderly might have difficulties using a normal interface, so an easy interface especially designed for them could be preferable. Using this app, elderly will be helped to do their easy exercises to stay in motion.	that helps working people move more and activates them to move more at for example a gym. The app will help them to be more open to exercise	An App that finds hotspots, and recommended walking routes through a city to explore.	App that athletes can keep up their food intake per day and see if they get the right vitamins etc.
Robot	Robot that shows students how to do exercises. Or tell them if they do it right.	A robot could be used to help elderly in their everyday movements.	A robot that supports working people to organize their sports activities	A robot that is leading a city tour	A robot that helps them with doing certain exercises and improves their form.
AI	Ai giving students the perfect training plan. And advise them something specific.	Ai could help elderly by giving them also their best training plan.	AI that is trained to give them the best training plan and can analyze movement through camera to make sure that the movement is performed correctly to diminish the chances of injuries	Trained Ai giving a personalized explore plan that advises tourists on things to do on vacation	AI could give them training schemes that are specific to their body and preferences etc.
Sensors	Sensors sensing if you do the right movement for climbing or starting a new sport. This way you can immediately see whether you are on the right track.	Sensors can detect whether elderly do their movements right for training.	Sensors could be used to check and correct the movement of the person. As well as the identify if the movement has been done the most optimal.	Sensors on a plane that helps tourists with their sitting/sleeping form. And helps them sit in the correct position.	Sensors keep up with their movement when they are doing an exercise. With this information you can improve their form and see if they need medical attention or concentrate on a specific part of their training.
Audio & Visual Technology	Students get to learn about the new sport through audio work.	Audio or Visual Technologies could help elderly give them precise instructions to move in a certain way that is safe for them, so they would not risk any accidents.	This can be used to make the work out more enjoyable through for example gamification.	Guided Audio Tour through a city	With audio and visual technology you can stimulate a specific situation so that they can train in this simulation without needing, for example, extra teammates or an opponent.

Figure 1: in this martrix the 5 by 5 matrix is visible in which we used to find interesting solutions to problems

Appendix 2: 50 concept ideas

App as a Technology

1) Dynamic scheduling for workouts, based on daily activities and lifestyle
2) AI progression planner, AI generating a progression planner, avoiding overloading
3) Workout planning, building efficient sessions into busy schedules
4) Pain-Point avoidance, suggesting workouts and exercises that minimizes stress on previously injured areas
5) Real time form checker, using phones to scan and analyze the form during workouts
6) Warning from bad postures during workout
7) Virtual coach, getting instant tips through headphones (could also be combined with the real time form checker)
8) Feedback after a session, getting detailed feedback after the workout about the form and movement
9) Using AI that detects struggle during exercises → which can later suggest different levels for example
10) A recovery timer could be added in the app as well, this way sporters know exactly how long the most optimal time of rest is
11) Also a rewarding system of maintaining safe practice/workouts could be added
12) Pressure sensor feedback, alerting people based on the grip/pressure during lifts
13) The app could also be connected to a wearable, like a wristband or an (apple) sports watch. This way vibrations could also deliver some message to the user
14) Specific features for one of the most injury-prone workout activities
15) A lift calculator, calculating the right amount of lifts that can be done with a specific weight
16) Education system within the app, educating on common causes of gym injuries
17) Visual guide video, a guide to every gym device
18) Safe gym quiz, making a quiz to test people on their knowledge, within the app, ensuring that afterwards people are knowledgeable enough to use the right exercises
19) Video tutorials in the app for every exercise
20) Keeping track of past injuries in the app and also adding the recovery journey
21) Buddy mode, keeping track of progress of your friend and being able to also participate in safe challenges

Robot as a technology

22) A robot that tells/shows people how to do a workout in the 'right' way
23) A robot interfering a possible dangerous exercise, to eliminate any further damages
24) A robot giving people a personal scheduled workout routine, like a coach
25) A robot correcting the form in real-time during exercises
26) A robot automatically balancing out unevenly loaded weights
27) A robot manually changing the weight for a certain person, this way injuries from putting the weight on and off are eliminated
28) A robot helping visually impaired people to complete the exercises correctly and with some assistance

AI as a technology

29) AI tracking user movements with a camera for direct feedback

30) Overlaying the posture on a monitor with a visual body of the perfect form, this way people can correct themselves, while doing the workout
31) Recommending safer and more efficient incremental training for each individual, based on their persona, body and lifestyle
32) Analyzing post workout to give the user a clear feedback on the session and on what they can improve next time to maximize time, result and safety

Sensors as a technology

33) Using cameras and sensors to track the full body movements in three dimensions
34) Vibrating to signal warnings for posture correctness or completion of workouts either with a wristband or a belt or some other gadget or wearable
35) Playing tones or instructions when the user deviates from the norm
36) Combining AI with sensors to predict risks of possible injuries
37) Sensors with incorporated lights, showing possible steps on machines in the gym
38) Sensors with colour codes, alerting users from good health rates with green lights and bad health rates with red lights
39) Sensors attached to the body, measuring the amount of breathing and the pulse, combined with an app visualizing a curve and through this create a schedule plan
40) Identifying uneven pressure distribution on lifting weights or squatting
41) Sensors monitoring grip strength, warning against improper holding techniques

Audio and Visual Technology

42) Vocal feedback, like an audio trainer based on real-time performance
43) Counting the repetition of workouts out loud through headphones or another speaker to make it easier for users to rely on the count so they can fully concentrate on the workout rather than having to count as well.
44) A voice assistant suggests adjusting the intensity of a workout or lower the intensity level in other cases.
45) Turning a workout into an interactive gamification
46) Making an interactive safety-focused challenge in a gym space
47) Using VR to mark and visually demonstrate possible risk spots for injuries.
48) Colour coded movement guide, green for correct postures and workouts, while red would be for incorrect postures and workouts
49) Showing users what muscles are being trained at the real time moment. This way people know exactly what they train and do not train one muscle group overly intensive
50) A monitor incorporated with an app that tracks when the user has reached its goal for the day for one specific part of the body. This way they can move on to another workout

Appendix 3: Interview questions

Interview questions

1. Hoe heet je? / What is your name?
2. Hoe oud ben je? / How old are you?
3. Hoe vaak ga je naar de sportschool? / How often do you go to the gym?
4. Wat voor oefeningen doe je in de sportschool? / What/How do you train in the gym?
5. Wat doe je in het dagelijks leven?
   (studie/sport/werk/hobbies?)
6. Hoe heb je geleerd om de (gewichtheffen) oefeningen die je uitvoert?
7. Heb je weleens een blessure gehad en kun je er meer over vertellen?
   a. (Waar?, hoe opgelopen? en hoe lang last?)
   b. (Hebben ze er wat van geleerd?)
8. in wat voor setting, met vrienden of alleen sport je vaak?
   a. wordt er weleens feedback gegeven over je vorm tijdens het trainen?
      i. Kan je meer over vertellen over hoe je je erbij voelt?
   b. Hoe zou je het prettig vinden om feedback over je vorm tijdens het trainen krijgen?
   c. Verandert het gewicht wat je pakt tijdens het trainen wanneer je met vrienden traint?
      i. hoe verandert het gewicht dan?
      ii. Waarom verandert het gewicht?
9. Weet je wat AI is?
10. Hoe hoog is je vertrouwensgevoel bij AI
    a. Wat maakt dat jij de AI (niet) vertrouwt?
11. Hoe zou jij je voelen bij het gebruik van een AI app op de telefoon die via de camera jouw aanwijzing geeft tijdens het trainen om je vorm te verbeteren?
12. Hoe zou jij je voelen bij het gebruik van een AI app op de telefoon om je trainingsschema en leven te combineren?
13. Hoe denk jij dat AI jou kan helpen bij het sporten en het voorkomen van blessures?

Appendix 4: Summary of the conducted interviews

Interview 1:
To summarize, participant 1 sports around 9+ hours a week. He does kickboxing, football, running and goes to the gym. He also works at least 12 hours in the kitchen. He is still a student which is not physically demanding. He learned to do the gym exercises from a friend that used to be in the military and likes sports. This friend also helped him make a training schedule. This friend kept the sport and life outside separate to prohibit overload of exercise. He does not do ego lifting since he is very strict with his schedule. He would use the app and thought a combination with a shirt that tracks body position would be nice. When he goes to the gym he goes 50/50 with friends.With friends they correct and encourage each other. When correcting the way is important. He gave an example that the best way would be: nice, but maybe next time do this because of this and that. Not to do, why are you doing this? This is not the best way, you should be doing this and that.

Interview 2:
To summarize, participant plays 2 sports around nine hours spread over 6 times a week. He does football and goes to the gym. He does a physical job, cleaning, but that is not very intense. According to him it does not influence his sporting abilities. He is still a student which is not physically demanding. He learned to do the exercises from his older brother and youtube. He goes to the gym preferably with his friends. When he goes to the gym with friends he does take heavier weights for some exercises. This is because he can be spotted and finds it unsafe to do this on his own. When he goes with friends they do correct each other on the execution of exercises. He prefers to be corrected in a way that enables him to grow. He also does not like it when the correction is done through a long story. Short and upbuilding is the best according to him. He does trust AI but questions how the database is built up. He would use the AI correction assistant only when he has to go to the physiotherapist. As well as the addition of the schedule assist.

Interview 3:
To summarize, participant 3 uses his house gym five times a week. The rest days are dependent on the schedule and how the participant feels physically that day. Usually the participant's current gym sessions are short (30-45 minutes) but intense, this is due to an only limited amount of time, as the participant's day to day life is busy. The training schedule is never the same, it gets mixed up with lots of variation, as you want to increase the stimulation of the muscle, so always give it a new stimulus. Amongst the muscle group areas are chest and triceps, back and biceps, legs, shoulders and repeat. Common exercises are chest press, push ups, lateral rows, deadlifts, squats, overhead press and many more… . The participant got taught how to do most exercises by his dad at first and for the perfect workout routine he followed a very profound coach on youtube, that was coaching the New York Mets (baseball team) for many years. During the journey of hitting the gym the participant has encountered a couple of injuries. Among them are hurting the shoulder and back through improper posture. The recovery time of this has been nagging as the pain sometimes still reappears depending on the held posture during a workout. From this the participant has learnt the value of proper warming up, stretching and recovery. When hitting the gym the participant prefers to do so on his own, as the workout gets done faster this way and the concentration level is higher. Even though friends do motivate him as well. He always uses the same weight. For corrections

about the posture the participant was very well receptive, as long as it is meant with good intentions. When getting feedback the participant thinks that an actual coach is best, as humans can view your posture from any angle, unlike an camera being stative at one location. The trust level in AI is there up until a certain level. The participant thinks that if he had an AI that would aid him with detection of good and bad posture, this would maximize the effectiveness of the workout and prevent injuries.

Interview 4:
To summarize, participant 4 is going to the gym 4 to 5 times a week. They use free weights but also machines. It differs per day what group muscles they train. He does go to work but it is not high intensity for the body, they work behind a laptop most of the time. He learned to do exercise mostly from video's (from tiktok). He goes to the gym alone but sometimes with friends. When they train together they do not really give feedback to each other. He also prefers not to get feedback from his friends, mainly because he already knows the exercises that he does very well. He does not change his weight when he trains with his friends. He does trust AI but thinks that a personal trainer is better to give advice regarding how to do an exercise correctly. He would find it useful when AI gives a personal plan, with food for example. Especially when people are just starting going to the gym and don't know what to do.

Appendix 5: Interview questions lo-fi testing

1. Can you do a deadlift?

2. What do you think about the persons form during the exercise ?

3. What do you think of the visual feedback that is given?

4. Would you be able to improve your form based upon the given feedback?

5. Can you elaborate ?

Appendix 6: Interview answers lofi prototype

Participant 1:

-Participant 1 has never tried a deadlift
-Person in video keeps touching the green line, so participant 1 assume she is doing it right
- Visual feedback: participant 1 thinks: green line is correct, red line is wrong. Standing is okay but its red? How she, person in video, moves: blue, green line goes with (correct position), and if its wrong then its red.
-Keep colourblind into account
-If lines are understandable/explained, then yes. Not now.

Participant 2:

- Yes, has done it but not very often. Participant 2 gyms alone, so participant 2 would need a spotter and thats why participant 2 doesnt do it.
- Iemand die een deadlift doet, ik zou nooit een deadlift zo snel doen. Participant 2 ziet de lijntjes wel maar niet duidelijk waar het voor is. Waar de vorm voor is en waar het voor zou moeten zijn
- Lijkt participant 2 handig om deze visuele feedback te krijgen. Het is niet heel duidelijk hoe de groen/rood lijnen voorstellen, meer context is nodig
- ja het zou kunnen helpen, vooral vanaf de zijkant kunnen zien is handig
- leuk idee!
- tip: geef meer context bij vragen

Particpant 3:

- Cannot do a deadlift
- I dont know, assume the green is correct and red is what they are doing. Then their form is pretty good
- Its a little bit confusing without any context, you would think red is wrong and not just her form and green is correct
- I think so, also needed where to put feet and how far apart they should be

Participant 4:

- Participant 4 can do a deadlift.
- Its hard to say considering the size of the plates, in my opinion why not, too much hip pinch, should go more through knees
- Depends on the purpose of the lines, lines should move with, is confused
- Idea of the lines is a good idea but needs to be interactive, more updates within time frame
    + perspective should be more from the side, also from the back for hip stabilization (left hip/right hip should be aligned before the deadlift)

Appendix 7 : Hi-Fi user testing form Group A (without video)

1. Name ?
2. Gender? (male/female/other)
3. Was it clear what to do when seeing the installation ?
4. How did you feel about the installation?
5. Did the LEDs grab your attention when you were doing the exercise ?
6. How did you feel about the colour of the LEDs?
7. What did you think of the intensity of the LEDs
8. Could you see yourself clearly on the screen (yes/no)
9. How do you feel about the angle/position of the camera?
10. How do you feel about seeing yourself on a screen while doing the exercise?
11. How did you interpret the colour of the lines on the screen?
12. What did you think was the purpose of the different lines?
13. How did you think this installation can help you ?
14. Would you like to add something?

Appendix 7b : Answers Hi-Fi user testing form Group A (without video)

Participant 1:

3. Yes
4. Very cool, but I dont know what it actually does.
5. Yes
6. Perfect placement for grabbing attention
7. Very bright, got scared a bit.
8. yes
9. ideal
10. I realized how bad my posture is
11. I dont remember the colour
12. Shows the overall posture.
13.Get a better deadlift posture
14.The system could tell me what and how to improve.

Participant 2

3. yes
4. I was a little bit confused about the feedback
5. yes
6. they were perceived functional
7. good!
8.Yes
9. alright. if software/work effort allows it, perhaps consider rearview
10. aware
11. a little bit confused
12. initial angle and suggested angle
13. I would consider it useful for the warming-up
14.great job! I hope the information helps

participant 3

3. Yes
4. I feel like it was clear to do a deadlift provided the information provided. But i did not understand what the main purpose of it was other then that the computer screen served as a mirror. Overall i was not super excited about it
5. Yes
6. It depends on the purpose of them. If they were telling me that i was doing something wrong they should have been red but i'm not sure of the purpose
7. It was bright enough
8. Yes
9. Good

10. The screen was a little too low on the floor which impacted my form
11. Those were nice to look at. Because i could see when my back would be too round.
12. To show the anatomy of the body clearly and indicate when bodies form is not good or good
13. It could help with deaflifts for beginners mostly. Maybe implementing it with audio cues or vibrations on the body would be more interesting for me too use.
14. There could be some instructions and more background info on the system given as there was context missing. But i like how it is programmed and that it is working!

Appendix 8 : Hi-Fi user testing form Group B (with video)

1. name?
2. Gender? (male/female/other)
3. Age?
4. What did you like and don’t like about the instruction video?
5. What do you think of the voice speaking in the video?
6. Was the instruction video clear? (yes/no)
7. Was it clear how to use the installation after watching the video. Explain your answer.
8. How did you feel about the lights? Were they helpful, distracting etc.
9. How did you feel about the colour of the LEDs
10. What did you think of the intensity of the LEDs?
11. Could you see yourself clearly on the screen? (yes/no)
12. How do you feel about the angle/position of the camera?
13. How do you feel about seeing yourself on a screen while doing the exercise
14. How did you interpret the colour of the lines on the screen ?
15. What do you think was the purpose of the different lines?
16. How do you think this installation can help you?
17. Would you like to add something?

Appendix 8b : Answers Hi-Fi user testing form Group B (with video)

Participant 1
4. i liked the visuals, it was comprehensive and easy to understand
5. it was easy to understand no complaints
6. yes
7. yes, the screen was a bit small in the instillation making things hard to see but other than that it was easy to understand
8.they were helpful, i knew that if they were on i had to adjust my position
9. maybe make them red to show that its bad rather than just white
10.the intensity was good, bright enough to draw attention without blinding the user
11. yes
12. good angle
13. a bit weird but also quite helpful
14. i liked that the correct position was shown in green, but maybe make the lines green if you are doing it correct or red if you arent
15. to show how my back is postioned
16. it helped me realize i have to keep a straight back and lean less forward\
17. no

Participant 2
4. It was clear and short, which is good
5. Clearly understandable
6. yes
7. It was clear since it was simple and easy to understand
8. The light made it easier to understand
9. Good
10.Little too intense
11. yes
12. good
13. It's really nice
14.blue bad, green good
15. guide me in the right direction
16.Yes
17.Cool project

Participant 3
4. It gave me a good vibe to start using the product. IT felt reliable.
5. Robotic, unpersonal
6. yes

7. The installation of the product was clear. However, as a novice in deadlifting, it did not help me into learning how a deadlift works. It works fairly good when you finally figured out how the deadlift works though.
8. The green line was enough of an indication that I was doing it wrong. The line itself gave me a nudge to go to the specific part. The LEDs distracted me.
9. Fine.
10. Too much.
11. yes
12. The screen was quite low, so I had to look down.
13. As long as I'm not looking straight to my face, it feels more than fine.
14.I would assume my own line would change red, when I did something wrong.
15. blue, my own posture. Green, the projected posture
16. It helped me get insights on deadlifting.
17. It overall looks very neat and professional!

Participant 4
4. I liked how it was short and clear. Maybe the instruction video could include pictures of the actual setup?
5. The voice sounded a little bit robotic to me
6. Yes
7. Yes
8. They didn't add too much for me personally because I was already focused on the screen. But they did give a clear yes or no on if my form was good or not.
9. Maybe a colour change would be easier on the eyes instead of on/off.
10. It was a little bright
11. yes
12. Good, maybe multiple cameras would present more data, but for one camera this is a good position.
13. feels weird, but not bad I think.
14. as the instruction video told me. Blue was me, green was ideal.
15. see previous answer
16.it could help me maintain safe form.
17. -op dit moment is hij niet fullscreen, niet zo makkelijk
-lampjes van kleur veranderen miss ipv aan/uit is miss rustiger voor je ogen
-weet niet of ik het altijd zou blijven gebruiken
-als alleen met telefoon en camera kon gebruiken, was het erg makkelijk
-handig dat je van de zijkant was

Appendix 9 : consent form Hi-Fi user testing

1. Name
2. I have read and understood the study information dated 17/01/2025, or it has been read to me. I have been able to ask questions about the study and my questions have been answered to my satisfaction. (yes/no)
3. I consent voluntarily to be a participant in this study and understand that I can refuse to answer questions and I can withdraw from the study at any time, without having to give a reason.(yes/no)
4. I understand that my abilities are not being tested through this study, only the product is being tested. (yes/no)
5. I understand that I will be interviewed after the user test and that my answers will be recorded. (yes/no)
6. I understand that the information I provide will be used for the project report. (yes/no)
7. I understand that personal information collected about me that can identify me, such as [e.g. my name], will not be shared beyond the study team. (yes/no)

Appendix 10 : Deadlift corrector script

```
import mediapipe as mp
import numpy as np
import math
import cv2
import serial
import time

#used to set up serial connection
arduino_port = 'COM4'  # Change this to your Arduino's COM port
baud_rate = 9600  # Baud rate, should match the Arduino's baud rate

#set upserial connection
ser = serial.Serial(arduino_port, baud_rate)
time.sleep(2)

#desired size of the window
window_width = 1920
```

```
window_height = 1080

# Initialize Mediapipe Pose
mp_pose = mp.solutions.pose

# colors
blue = (255, 0, 0)
white = (255, 255, 255)
green = (0, 255, 0)
red= (0, 0, 255)
# Phase flags
start_position_1 = False
start_position_2 = False
mid_lift = False
finished_lift_1 = False
finished_lift_2 = False
no_lift = False  # Default value

# Memory flags
memory_start_position_1 = start_position_1
memory_start_position_2 = start_position_2
memory_mid_lift = mid_lift
memory_finished_lift_1 = finished_lift_1
memory_no_lift = True


# Function to calculate angle between three points
def calculate_angle(a, b, c):
    a = np.array(a)  # First point
    b = np.array(b)  # Mid point
    c = np.array(c)  # End point

    radians = np.arctan2(c[1] - b[1], c[0] - b[0]) - np.arctan2(a[1] - b[1], a[0] - b[0])
    angle = np.abs(radians * 180.0 / np.pi)

    if angle > 180.0:
        angle = 360 - angle

    return angle


# Function to calculate distance between two points
def distance(position_one, position_two):
    x_1, y_1 = position_one
    x_2, y_2 = position_two
    dist = math.sqrt((x_2 - x_1) ** 2 + (y_2 - y_1) ** 2)
    return dist
```

```
# Function to check the deadlift phase based on angles
def check_deadlift_phase(torso_angle, hip_angle, knee_angle):
    global start_position_1, start_position_2, mid_lift, finished_lift_1, finished_lift_2, no_lift
    global memory_start_position_1, memory_start_position_2, memory_mid_lift,
memory_finished_lift_1, memory_no_lift

    # Start lifting phase 1
    if 136 <= hip_angle <= 170 and memory_no_lift:
        no_lift = False
        start_position_1 = True
        memory_start_position_1 = True

    # Start lifting phase 2
    elif 103 <= hip_angle <= 135 and memory_start_position_1:
        start_position_1 = False
        memory_no_lift = False
        start_position_2 = True
        memory_start_position_2 = True

    # Mid lift
    elif 65 <= hip_angle <= 103 and memory_start_position_2:
        start_position_2 = False
        memory_start_position_1 = False
        mid_lift = True
        memory_mid_lift = True

    # End lift phase 1
    elif 103 <= hip_angle <= 135 and memory_mid_lift:
        mid_lift = False
        memory_start_position_2 = False

        finished_lift_1 = True
        memory_finished_lift_1 = True

    # End lift phase 2
    elif 140 <= hip_angle <= 170 and memory_finished_lift_1:
        finished_lift_1 = False
        memory_mid_lift = False

        finished_lift_2 = True

    # No lift
    elif hip_angle >= 150 and knee_angle >= 150:
        no_lift = True
        memory_no_lift = True
        memory_finished_lift_1 = False
        finished_lift_2 = False
```

```
    return start_position_1, start_position_2, mid_lift, finished_lift_1, finished_lift_2, no_lift


# Returns the perfect angles per phase
def perfectAngle(phase_b1, phase_b2, phase_m, phase_e1, phase_e2):
    if phase_b1:
        pAnkle = 90
        pKnee = 75
        pHip = 70
        pNeck = 65
    elif phase_b2:
        pAnkle = 90
        pKnee = 71
        pHip = 50
        pNeck = 55
    elif phase_m:
        pAnkle = 90
        pKnee = 67
        pHip = 25
        pNeck = 35
    elif phase_e1:
        pAnkle = 90
        pKnee = 71
        pHip = 50
        pNeck = 55
    elif phase_e2:
        pAnkle = 90
        pKnee = 75
        pHip = 70
        pNeck = 65
    else:
        pAnkle = 83
        pKnee = 80
        pHip = 92
        pNeck = 85
    return pAnkle, pKnee, pHip, pNeck


# Function to calculate optimal position
def perfectPositions(ankle, ankleAngle, shin_length, kneeAngle, thigh_length, hipAngle, torso_length, neckAngle, neck_length):
    ankle_x, ankle_y = ankle

    # Convert angles from degrees to radians
    ankle_angle_rad = math.radians(ankleAngle)
    knee_angle_rad = math.radians(kneeAngle)
    hip_angle_rad = math.radians(hipAngle)
```

```
    torso_angle_rad = math.radians(neckAngle)

    # Calculate the knee position based on the ankle and shin length using trigonometry
    x_knee = ankle_x - shin_length * math.cos(ankle_angle_rad)
    y_knee = ankle_y - shin_length * math.sin(ankle_angle_rad)

    # Calculate the hip position based on the knee position, thigh length, and knee angle
    x_hip = x_knee + thigh_length * math.cos(knee_angle_rad)
    y_hip = y_knee - thigh_length * math.sin(knee_angle_rad)

    # Calculate the torso position based on the hip position, torso length, and torso angle
    x_torso = x_hip - torso_length * math.cos(hip_angle_rad)
    y_torso = y_hip - torso_length * math.sin(hip_angle_rad)

    # Calculate the torso position based on the hip position, torso length, and torso angle
    x_neck = x_torso - neck_length * math.cos(torso_angle_rad)
    y_neck = y_torso - neck_length * math.sin(torso_angle_rad)

    return x_knee, y_knee, x_hip, y_hip, x_torso, y_torso, x_neck, y_neck


# Check if the form is good based on the angles
def goodForm(knee, perfectKnee, hip, perfectHip, neck, perfectNeck, midLift):
    form_hip = 0
    # calculates the difference between the perfect angle and live angle per joint
    if hip > 90 or neck > 90:
        form_hip = abs(hip - perfectHip-90)
        form_neck = abs(knee - perfectNeck -90)
    else:
        form_hip = abs(hip -perfect_hip)
        form_neck = abs(knee - perfectNeck)
    form_knee = abs(knee - perfectKnee -90 )

    #determines the form of the person
    if midLift and form_knee < 20 and form_hip < 20 and 12 < form_neck <40 :
        goodPosition = True
    elif not midLift and form_knee< 15 and form_hip < 40 and 12 < form_neck <25:
        goodPosition = True
    else:
        goodPosition = False

    return goodPosition


# Set up video capture and width, height
cap = cv2.VideoCapture(1)
width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
```

```
cv2.namedWindow('Mediapipe Feed', cv2.WND_PROP_FULLSCREEN)
cv2.setWindowProperty('Mediapipe Feed', cv2.WND_PROP_FULLSCREEN,
cv2.WINDOW_FULLSCREEN)

# Setup Mediapipe Pose instance
with mp_pose.Pose(min_detection_confidence=0.5, min_tracking_confidence=0.5) as pose:
    while cap.isOpened():
        ret, frame = cap.read()

        # Recolor image to RGB for Mediapipe processing
        image = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
        image.flags.writeable = False

        # Process the frame to get pose landmarks
        results = pose.process(image)

        # Recolor the image back to BGR for OpenCV display
        image.flags.writeable = True
        image = cv2.cvtColor(image, cv2.COLOR_RGB2BGR)

        try:
            if results.pose_landmarks:
                # Extract landmarks for key points
                landmarks = results.pose_landmarks.landmark
                earL = [landmarks[mp_pose.PoseLandmark.LEFT_EAR.value].x,
                        landmarks[mp_pose.PoseLandmark.LEFT_EAR.value].y,
                        landmarks[mp_pose.PoseLandmark.LEFT_EAR.value].z]
                shoulderL = [landmarks[mp_pose.PoseLandmark.LEFT_SHOULDER.value].x,
                             landmarks[mp_pose.PoseLandmark.LEFT_SHOULDER.value].y,
                             landmarks[mp_pose.PoseLandmark.LEFT_SHOULDER.value].z]
                hipL = [landmarks[mp_pose.PoseLandmark.LEFT_HIP.value].x,
                        landmarks[mp_pose.PoseLandmark.LEFT_HIP.value].y,
                        landmarks[mp_pose.PoseLandmark.LEFT_HIP.value].z]
                kneeL = [landmarks[mp_pose.PoseLandmark.LEFT_KNEE.value].x,
                         landmarks[mp_pose.PoseLandmark.LEFT_KNEE.value].y,
                         landmarks[mp_pose.PoseLandmark.LEFT_KNEE.value].z]
                ankleL = [landmarks[mp_pose.PoseLandmark.LEFT_ANKLE.value].x,
                          landmarks[mp_pose.PoseLandmark.LEFT_ANKLE.value].y,
                          landmarks[mp_pose.PoseLandmark.LEFT_ANKLE.value].z]
                heelL = [landmarks[mp_pose.PoseLandmark.LEFT_HEEL.value].x,
                         landmarks[mp_pose.PoseLandmark.LEFT_HEEL.value].y,
                         landmarks[mp_pose.PoseLandmark.LEFT_HEEL.value].z]
                toesL = [landmarks[mp_pose.PoseLandmark.LEFT_FOOT_INDEX.value].x,
                         landmarks[mp_pose.PoseLandmark.LEFT_FOOT_INDEX.value].y,
                         landmarks[mp_pose.PoseLandmark.LEFT_FOOT_INDEX.value].z]

                # Convert normalized landmark coordinates to pixel coordinates
```

```
            earL_pixel = tuple(np.multiply(earL[:2], [width, height]).astype(int))  # Use only x and y
            shoulderL_pixel = tuple(np.multiply(shoulderL[:2], [width, height]).astype(int))  # Use only x and y
            hipL_pixel = tuple(np.multiply(hipL[:2], [width, height]).astype(int))  # Use only x and y
            kneeL_pixel = tuple(np.multiply(kneeL[:2], [width, height]).astype(int))  # Use only x and y
            ankleL_pixel = tuple(np.multiply(ankleL[:2], [width, height]).astype(int))  # Use only x and y
            heelL_pixel = tuple(np.multiply(heelL[:2], [width, height]).astype(int))  # Use only x and y
            toesL_pixel = tuple(np.multiply(toesL[:2], [width, height]).astype(int))  # Use only x and y

            # Calculate angles
            torso_inclination = np.rint(calculate_angle(earL[:2], shoulderL[:2], hipL[:2]))
            hipL_inclination = np.rint(calculate_angle(shoulderL[:2], hipL[:2], kneeL[:2]))
            kneeL_inclination = np.rint(calculate_angle(hipL[:2], kneeL[:2], heelL[:2]))
            ankleL_inclination = np.rint(calculate_angle(kneeL[:2], heelL[:2], toesL[:2]))

            # Calculates the distance of shin and thigh
            shin_dist = distance(ankleL[:2], kneeL[:2])
            thigh_dist = distance(kneeL[:2], hipL[:2])
            torso_dist = distance(hipL[:2], shoulderL[:2])
            neck_dist = distance(shoulderL[:2], earL[:2])
            # Find movement phase
            start_position_1, start_position_2, mid_lift, finished_lift_1, finished_lift_2, no_lift = check_deadlift_phase(
                torso_inclination, hipL_inclination, kneeL_inclination)

            # Find the desired angle based on the current movement phase
            perfect_ankle, perfect_knee, perfect_hip, perfect_neck = perfectAngle(start_position_1, start_position_2, mid_lift,
                                                                                  finished_lift_1, finished_lift_2)

            # Find perfect coordinates of the knee, hip, torso based on the phase
            knee_x, knee_y, hip_x, hip_y, torso_x, torso_y, neck_x, neck_y = perfectPositions(ankleL[:2], perfect_ankle, shin_dist,
                                                                                              perfect_knee, thigh_dist, perfect_hip,
                                                                                              torso_dist, perfect_neck, neck_dist )

            goodform = goodForm(kneeL_inclination, perfect_knee, hipL_inclination, perfect_hip, torso_inclination, perfect_neck, mid_lift)

            # Convert normalized positions to pixel positions
            knee_pixel = (int(knee_x * width), int(knee_y * height))
            hip_pixel = (int(hip_x * width), int(hip_y * height))
```

```
torso_pixel = (int(torso_x * width), int(torso_y * height))
neck_pixel = (int(neck_x * width), int(neck_y * height))

if goodform:
# Draw all of the landmarks as circles
    cv2.circle(image, earL_pixel, 7, white, -1)
    cv2.circle(image, shoulderL_pixel, 7, white, -1)
    cv2.circle(image, hipL_pixel, 7, white, -1)
    cv2.circle(image, kneeL_pixel, 7, white, -1)
    cv2.circle(image, ankleL_pixel, 7, white, -1)
    cv2.circle(image, heelL_pixel, 7, white, -1)

    # Draw line between the landmarks
    cv2.line(image, earL_pixel, shoulderL_pixel, blue, 2)
    cv2.line(image, shoulderL_pixel, hipL_pixel, blue, 2)
    cv2.line(image, hipL_pixel, kneeL_pixel, blue, 2)
    cv2.line(image, kneeL_pixel, ankleL_pixel, blue, 2)
    cv2.line(image, ankleL_pixel, heelL_pixel, blue, 2)
    cv2.line(image, heelL_pixel, toesL_pixel, blue, 2)

    #makes sure that the leds are turned of
    ser.write(b'T\n')
if not goodform:

    # Draw corrected circles at knee, hip, and torso positions
    cv2.circle(image, ankleL_pixel, 7, white, -1)
    cv2.circle(image, heelL_pixel, 7, white, -1)
    cv2.circle(image, knee_pixel, 7, white, -1)
    cv2.circle(image, hip_pixel, 7, white, -1)
    cv2.circle(image, torso_pixel, 7, white, -1)
    cv2.circle(image, neck_pixel, 7, white, -1)

    # Draw corrected lines connecting the knee, hip, and torso
    cv2.line(image, toesL_pixel, heelL_pixel, green, 2)
    cv2.line(image, heelL_pixel, ankleL_pixel, green, 2)
    cv2.line(image, ankleL_pixel, knee_pixel, green, 2)
    cv2.line(image, knee_pixel, hip_pixel, green, 2)
    cv2.line(image, hip_pixel, torso_pixel, green, 2)
    cv2.line(image, torso_pixel, neck_pixel, green, 2)

    #draw body lines
    cv2.circle(image, earL_pixel, 7, white, -1)
    cv2.circle(image, shoulderL_pixel, 7, white, -1)
    cv2.circle(image, hipL_pixel, 7, white, -1)
    cv2.circle(image, kneeL_pixel, 7, white, -1)
    cv2.circle(image, ankleL_pixel, 7, white, -1)
    cv2.circle(image, heelL_pixel, 7, white, -1)
```

```
            # Draw line between the landmarks
            cv2.line(image, earL_pixel, shoulderL_pixel, red, 2)
            cv2.line(image, shoulderL_pixel, hipL_pixel, red, 2)
            cv2.line(image, hipL_pixel, kneeL_pixel, red, 2)
            cv2.line(image, kneeL_pixel, ankleL_pixel, red, 2)
            cv2.line(image, ankleL_pixel, heelL_pixel, red, 2)
            cv2.line(image, heelL_pixel, toesL_pixel, red, 2)

            # Send signal to turn on LaEDS
            ser.write(b'F\n')
    except Exception as e:
        print(f"Error: {e}")

    # Show the updated image
    cv2.imshow('Mediapipe Feed', image)

    # Exit the loop if 'q' is pressed
    if cv2.waitKey(10) & 0xFF == ord('q'):
        ser.write(b'S\n')
        break

cap.release()
cv2.destroyAllWindows()
ser.close()
```

Appendix 11: Arduino script

```
int LEDs = 6;

void setup() {
  Serial.begin(9600);
  pinMode(LEDs, OUTPUT);

}
void loop() {
  if (Serial.available() > 0) {
    delay(10); // Add a short delay to allow for data stabilization
    String msg = Serial.readString();

    if (msg == "F") {
      analogWrite(LEDs, 30);
    } else if (msg == "T") {
      analogWrite(LEDs, 0);
    }
  }

}
```

Appendix 12: Fusion file screen encasing

LEDs screen encasing used for lasercutting: https://drive.google.com/file/d/1MH64gX7P-IynmvgqcOenLpLVZYNmOvHF/view?usp=sharing

Appendix 13: Videos used during Hi-Fi evaluation

Deadlift tutorial video:
https://drive.google.com/file/d/1qRofeOXDdEidyZCyHTATXixRv2_aoPIc/view?usp=sharing

LiveLift installation instruction video:
https://drive.google.com/file/d/1MeILs1p6GIv4Z2tZY9fMbs4fFM-UVaBr/view?usp=sharing